

# FoodBlock: A Secure and Cost-Optimal Framework for Online Food Ordering using Blockchain

Uday Mittal, Shivam Goyal, Eгна Praneeth Gummana, Vinay Chamola, Shivi Agarwal, and Trilok Mathur

**Abstract**—The internet has drastically changed trade and interpersonal interactions since the introduction of the pandemic. Food delivery services have expanded significantly in recent years. These services link customers with restaurants using the internet, enabling quick delivery of mobile meals. But as these businesses grow, dealing with problems like fraud and inefficiencies that may lower the quality of their services is more crucial than ever. This article lists the drawbacks of the existing food delivery services and recommends how to fix them. A blockchain-based architecture is also used to increase security and reliability, and an auction process encourages fairness and transparency for both riders and restaurants. We have developed an auction model where the riders can quote the price they estimate for a particular delivery. Additionally, a machine-predicted bid is also used to prevent the case where the riders would try to exploit the restaurants. This method leads to a better way of implementing online food delivery services, safeguarding the interests of customers, riders, and restaurants.

**Index Terms**—Food delivery services, Blockchain, Food-tech, Machine Learning, Auction

## I. INTRODUCTION

In recent years, online food purchasing services have expanded at an accelerated rate. Similar to how Uber provides a platform for drivers and customers to interact, these food delivery services are affiliated with riders who use an internet-based application to pick up food from a restaurant and deliver it to the customer's location. This alleviates the burden on restaurants that would otherwise be required to maintain their delivery fleet and enables them to offer food delivery services at minimal additional expense. It gratifies consumers by delivering food directly to their doorsteps and allows drivers to earn a substantial income while working for these companies. In addition to receiving commissions from restaurants for orders and serving as a marketing channel, online food delivery service providers have made a lot of money by accepting commissions from restaurants and acting as a marketing channel for restaurants. As with all services, these benefits are not flawless. Occasionally, these platforms have disappointed customers and restaurants due to fraudulent

customers and late deliveries. In addition, their systems have been unjust to riders by not distributing their orders evenly.

This paper uses blockchain technology [1] to address issues such as fraudulent clients and late deliveries before introducing an auction mechanism [2] for online food delivery platforms [3]. [Blockchain's numerous benefits have made it a game-changer in multiple industries.](#) It is well-suited for a future where users demand control and privacy that can only be provided by a distributed system [4]. Using blockchain, we can have riders submit bids for the prices at which they are willing to complete the delivery.

The restaurant conducts the auction and selects a suitable rider to complete the task based on a combination of compensation factors. With our proposed model, the restaurant will be able to generate greater profits, and the riders will earn more. Our model focuses predominantly on restaurants and riders as the primary entities [5]. The restaurants establish minimum and maximum food order bids. The restaurant then conducts an auction among the riders, during which the riders submit bids for the food order. The recommendations indicate how much the rider desires to be compensated for delivering the order successfully. The riders receive the machine-predicted proposal - a ballpark figure that the restaurant considers reasonable for that particular order - and then begin bidding on the food order. This bargaining occurs in three rounds [6]. The restaurant ultimately chooses the auction victor or rider to deliver the order. The decision is also contingent on the order's price. If the order cost is low, the restaurant can take a chance on a lower-rated rider; if the order cost is high, the restaurant should opt for a higher-rated rider who is more likely to keep the order secure and transport it quickly. The decision also hinges on the rider's rating and past performance.

[The following paper is organized in the following format. Section II deals with various previous works related to this topic. Then, in the next section III, the emergence of blockchain technology is described in synopsis format, along with various applications of the blockchain in the real world, especially related to this current work. In Section IV, the present-day food delivery systems have been described. In the further section V, various drawbacks of the existing system and the improvements it offers are discussed. We have developed the model in detail from section VI onwards, we have developed the model in detail. Sections VI, VII, VIII, and IX deal with it by proposing the model, describing the methodology, the auctioning model, and analyzing the model based on numbers \(numerical analysis\). Section X deals with the results obtained during the whole study, section XI deals](#)

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with various future researches, which can be done in a way to carry this work forward, and at last section XII provides the conclusion of the whole study.

Some of the significant contributions that the paper offers to the scientific world are

- Establishing that the decentralized system of food delivery is better than the traditional centralized system of food delivery
- The decentralized system prevents the exploitation of the riders as the riders could themselves choose whether to accept a particular delivery or not, in contrast to them being employees of the restaurants or the food delivery companies.
- Customers have a greater say in the delivery process, where they can democratically decide which rider to choose for their delivery and which not to choose.
- Increase in the riders' earnings and a significant decrease in the food delivery time.

## II. RELATED WORKS

There has been no prior research on auctions for the delivery of food orders. The prototype [9] was presented, demonstrating that customers could order from restaurants and pick up the order themselves if they wished to consume the food at home; the entire transaction would be conducted via telephone. The online food ordering industry [10] is expanding. Numerous applications have led people to order food from online food-demanding applications rather than dining in restaurants, as these applications offer substantial discounts to customers, and ordering food online does not require you to leave your home or pay service fees levied by restaurants. The authors of [7] describe the brute force method of online food ordering, in which consumers, riders, and restaurants use three servers to maintain their respective data. *Christidis et al.* [11] suggested an algorithm in which the restaurant's and customer's areas are linked with a geocode that reveals the location of both the restaurant and customer, and a random rider is chosen to deliver the order. Table I mentions some of the previous works in this area and their strengths and weaknesses.

## III. EMERGENCE OF BLOCKCHAIN TECHNOLOGY

Blockchain has been the game changer in many industries like car rental platform [12], crowdfunding outlet [13], credit scoring [14], [15], Industrial IoT [16], [17], [18], [19], agri-food supply chain management [20], healthcare [21], securing intelligent cities [22], IoT applications [23], [24], Food court payment [25], etc. Blockchain (*Hassija et al., 2019* [26]) is a technology that makes it possible to create a distributed, irreversible digital ledger that is decentralized and may be used to record transactions between several parties in a complex peer-to-peer network. A blockchain record cannot be changed backward without changing all previous transactions and gaining network agreement. The majority of online meal ordering services aim to increase rapid food delivery and revenue for caterers and restaurants (*Hassija et al., 2020* [27]). Although blockchain is often connected to cryptocurrency, it

is increasingly seen as a problem-solving tool that will boost productivity and delivery times. Blockchain networks on an industrial scale are starting to appear, even though blockchain technology is generally linked to cryptocurrencies like Bitcoin [28]. On the stock market, this new technology has already made a name for itself (*Bansal et al., 2019* [29]).

## IV. CURRENT SCENARIO

Typically, customers, restaurants, and drivers sign up for the systems now available for online food ordering. Restaurants must pay a registration fee, while consumers and riders may register for free. Several eateries are available to users, each with a menu, details, and reviews from previous customers. Customers choose the food they want when ordering, provide the delivery address, and then pay. An algorithm then looks for the next available rider to finish the delivery while transmitting the order information and the user's address to the restaurant of choice. The rider who accepts the delivery price is given the order, and the search is limited to a two-kilometer area around the business. The rider obtains the user's address and is responsible for timely delivery. Upon successful delivery, the restaurant gets 70% of the whole order amount, and the rider is compensated with a fee of around 10%. Additionally, riders may get prizes for extreme efforts throughout the month. This process is followed when a customer decides to make an online payment. When cash-on-delivery is desired, the rider must send the necessary funds to the eatery; all other stages remain the same [30] [31] [3]. Several issues with the current system are covered more deeply in the next section V.

## V. DRAWBACKS OF THE CURRENT SCENARIO AND THEIR IMPROVEMENTS

The improvements given will surely bring about a change in current online food ordering platforms.

### A. Fraud customers

Many customers are not the actual beneficiaries of the food, but they receive the food cheatingly. Such cases can be avoided using a one-time password (OTP) mechanism [32]. This mechanism will work in the following steps.

- The customer first places an order in the restaurant using the food delivery application or website. A rider would be assigned to deliver the food to the customer. Details of how the rider would be assigned will be discussed in later sections VI.
- A unique one-time password would be generated, which would be shared with the customer and the rider at the time of the delivery of the food.
- The customer would be bound to share that password with the rider as a verification measure, ensuring that the food is handed over to the right person.

Employing the OTP-based system could avoid instances of fraudulent deliveries and provide better customer service could be provided. The OTP could be generated using a random number generator function, which could handle a certain number of people at a time [33]. The process could

TABLE I: Related works on online food ordering

| References                     | Contributions                                                                                                                   | Strengths                                                     | Weaknesses                                            |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------|
| online food ordering           | Restaurants had their own riders to deliver the orders                                                                          | Restaurants had to deal with the riders                       | Restaurants who can't afford riders couldn't enter    |
| Scifo <i>et al.</i> [7]        | Restaurants had to telephonically call the riders informing them the details of the order                                       | Restaurants knew which rider to call                          | There were possibilities that rider denies the order  |
| Zhou and Han <i>et al.</i> [8] | Frameworks propose a solution in which the rider with highest rating near restaurant delivers the order                         | This system ensured faster delivery                           | Riders with low rating could not earn                 |
| Our proposed approach          | Proposed an auction for online food ordering platform where rider with best activity and less bid will get to deliver the order | Bigger profits for restaurants and better earnings for riders | Riders with less rating will get orders of less value |

be initiated once the rider is allocated to the request, but it would be better to share it at the time of delivery to avoid the cases of losing the OTP or leaking the OTP, etc.

### B. Late delivery

The window of time given to customers after purchase confirmation is often seen as an estimate and is thus liable to change. Some companies may not accept responsibility for orders that are delayed in transit. The customers are not satisfied since their meals were cold and delivered late. The shortage of orders placed in specific places and riders accessible citywide is to blame for these problems. An algorithm assigns a rider to delivery by giving preference to the rider who is closest to the consumer and has the highest rating from that customer. Technology issues are often blamed for delivery delays in today's corporate sector. But mistakes of this kind can't keep happening. The late delivery is likely due to the app or website's flawed algorithm. [Frequent tweaks to the algorithm are required to ensure on-time delivery and complete client satisfaction. In an ideal scenario, an algorithm would track down riders who have just been through the delivery zone, alert them to the order specifics, and set up a bidding platform where the business could offer a delivery price.](#) The rider whose bid is closest to the delivery's actual cost is chosen for the task. The firm uses an algorithm to determine which citywide riders have the fewest deliveries to do and then shares that information with them. This is done so that every rider receives the same daily pay.

## VI. PROPOSED MODEL

Most websites enabling users to purchase meals online employ an automated rider selection process. As part of the proposed strategy, our model selects riders apprised of the schedule using the K nearest neighbors method [34]. Each rider receives a machine-predicted bid that indicates the amount they should propose for the delivery based on their performance and merit. The riders then participate in a three-round bargaining process for the meal order [6]. The ratings and past versions of the riders are crucial in determining who will win the auction. After the auction [35], the restaurant receives all the riders' proposals and ratings. Based on this information, the restaurant chooses the rider to deliver the order. The restaurant may select a low-rated rider if the order is low-value. To ensure the safety of the order and expedite delivery to the customer, the restaurant prioritizes the selection of a rider with a high rating when the order's value is high [36]. Blockchain is the central point of communication for users, restaurants, and riders.

Blockchain-based transactions preserve privacy-preserving for all three parties involved. Users order food dishes through a DApp. Once a user places an order, a block is formed with the user ID, food item ID, user address, cost decided, and the restaurant ID. The auction happens over immutable blockchain transactions. This ensures high security in the bidding process. After the bidding ends, the restaurant chooses a rider. [A block is created on the blockchain with the food item ID, final cost to the rider, user cost, user ID, rider ID, restaurant ID, and user address.](#) The delivery agreements between the three parties are formed using smart contracts. When the user asserts the arrival of the ordered food item, the amount automatically debits from the user's account to the restaurant's and from the restaurant's account to the rider's [37]. [An outline of the proposed methodology is given in Figure 1 and 2.](#) The details of our proposed strategy are outlined in the following section.

### A. Nearest Riders

Selecting a predetermined number of local riders and informing them of the order is the first step in our proposed methodology. [It is done in two steps. Firstly, a two-kilometer radius is being constructed around the restaurant. Then, all the riders in that radius would be informed about the order and invited to participate in the auction process. If no rider is available in the restaurant's two-kilometer radius, then K Nearest Neighbour \[38\] would be used to identify the riders closest to the restaurant. A predetermined number of riders will be selected to participate in the auction procedure \[2\]. After selecting the rider, the anticipated arrival time of each rider at the restaurant will be calculated. The top riders will be selected after sorting the riders according to this determined time. This algorithm will only require a single iteration. Once identified, the closest K riders will immediately be informed of the order.](#)

### B. Final Rider Rating

[In our proposed model, the rating of the rider, obtained by delivering the previous orders, and the current customer rating, provided by the customer for the current delivery, will be considered when calculating the final rider rating. The customer rating is determined by calculating the weighted average of all the ratings the rider has received to that particular point. The customer rating that the rider has received for the most recent order will be provided the highest weightage and clubbed with the previous rider ratings, as per a method, which will be explained in the subsequent sections. The final rider rating will be one of the most important factors in the auction process.](#)

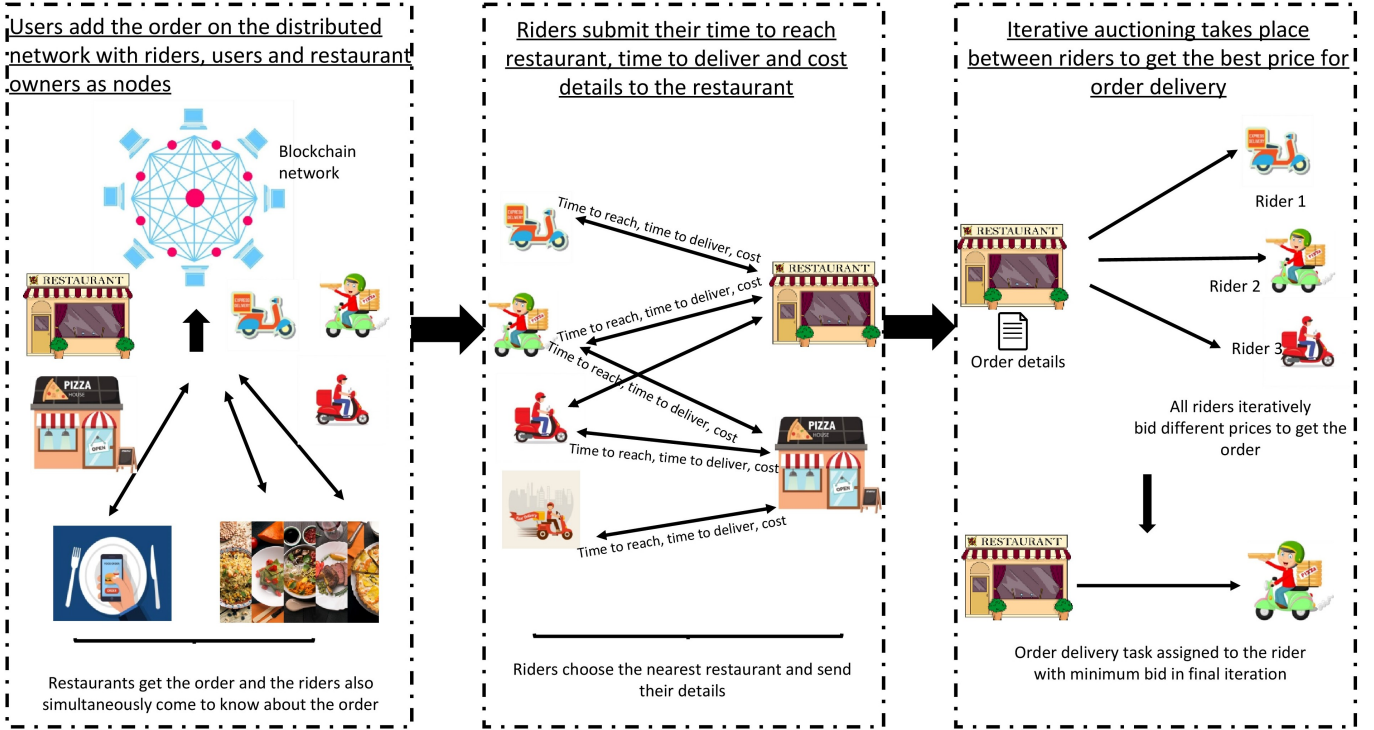


Fig. 1: Proposed model for online food ordering using blockchain.

### C. Bid Assignment

The device in our design provides each rider with an order with a maximum and minimum bid amount [39]. These figures are computed using two variables: distance and rider rating. However, the distance parameter remains unchanged for all the bid proposals.

When a user purchases a meal from the restaurant, the decentralized system considers the orders the riders delivered that day and the accompanying customer ratings. Due to the utilization of these two variables, every rider receives a table containing a machine-predicted bid. The machine-predicted bid is based on the cumulative rider rating. The cumulative rider rating is calculated using user feedback for all orders delivered until that time, which includes all the orders delivered that same day.

The machine prediction technique assigns weights to the parameters of orders delivered and user rating:  $w_1$  for the number of orders a rider delivered on a given day and  $w_2$  for the aggregate rating of the rider. The presence of weight  $w_2$  is essential to prevent the rider from the potential threat of some customers giving wrong ratings. Secondly, it also safeguards his interests because it may happen that someday, the rider's performance might be impaired due to any number of circumstances. Hence, weight  $w_2$  makes the whole process more humane [40].

### D. Algorithm

Multiple riders, dining establishments, and food orders comprise our system model. When a client orders from a restaurant listed on the online food delivery application, our algorithm is designed to identify the closest rider available

to the restaurant. The rider server, which provides access to the current locations of working riders, facilitates this inquiry. The server looks for riders within a two-kilometer radius of the restaurant.

In cases with insufficient riders within a two-kilometer radius, we provide a parameter  $k$  that specifies the minimum number of riders required for the delivery auction. Depending on the number of riders present in a city on a particular day, the value of  $k$  may vary between cities [41]. We require a sufficient number of riders to participate in the bidding process; accordingly, the extent of the space next to the establishment may vary based on the number of available riders [42].

After a sufficient number of riders have been located nearby, the algorithm determines the expected time required for each rider to reach the restaurant. This minimal time ranks the riders, with those nearest the restaurant listed first. The top  $k$  riders are then selected and apprised immediately of the order's details, including the products ordered, the restaurant's location, and the customer's address. Our general algorithmic framework implements the K Nearest Neighbour algorithm in this manner [43].

Before selecting  $k$  passengers, the dining establishment establishes a minimum and maximum offer range. The proposal amounts are then displayed to the riders. Within the restaurant's minimum and maximum bid range, the riders may place any proposal they choose [44].

Riders also receive a machine-predicted bid based on various variables, including their current customer rating, order history, and location. The average of all ratings the rider has received is used to determine the customer rating.



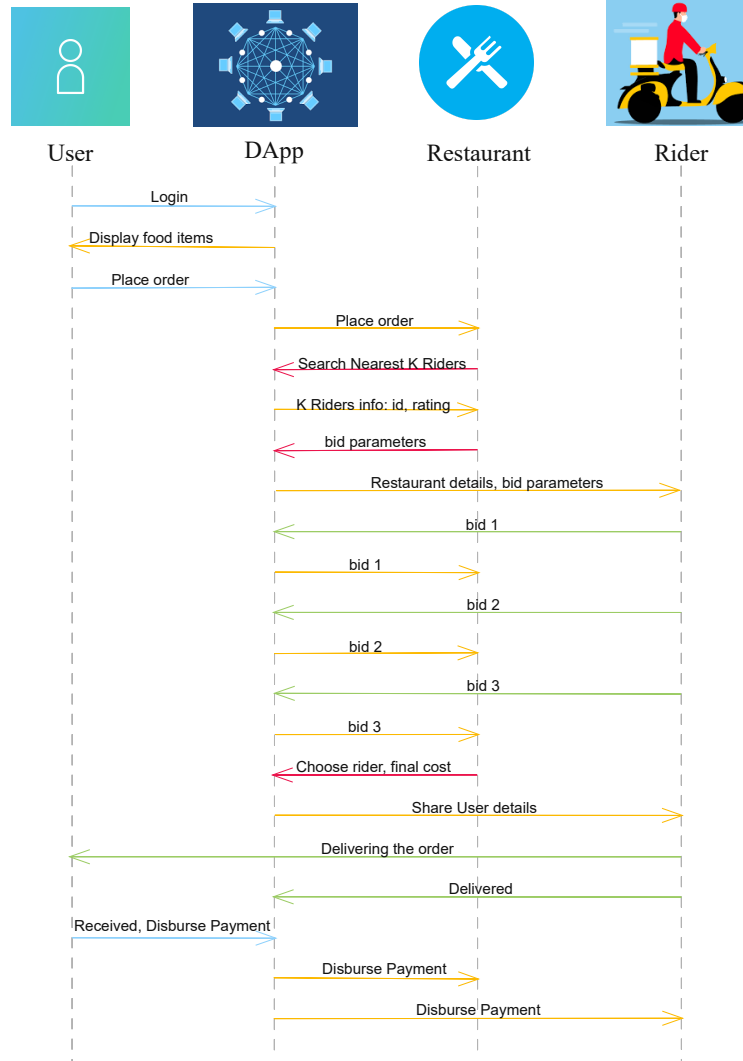


Fig. 2: Sequence Diagram for the online food delivery using blockchain.

After the machine-predicted offer has been determined, the riders can submit their proposals for the order. Each rider may submit their proposal, which must fall within the restaurant's established minimum and maximum bid range [45].

#### E. Rider Selection

All riders have received the machine-predicted bids based on their order history and customer ratings. To fulfill the order, they have submitted their proposals. The restaurant must now determine which rider will be responsible for delivering the food.

The restaurant should prioritize order delivery security over cost-cutting measures in making decisions. The restaurant may save money by selecting a rider with a low bid, but there is no guarantee that the client's order will be delivered safely. In contrast, a rider with a high bid could ensure the secure delivery of the customer's order, but it may not be cost-effective for the restaurant [46].

To overcome this problem, the restaurant must decide based on the order's cost. If the purchase is costly, it may be prudent to designate delivery to a rider with a high quote to ensure that it is delivered to the client securely. If the order's value

is minimal, the restaurant may assign the delivery to a rider with a lower offer to save money [47].

Cost considerations will ultimately influence the restaurant's selection of delivery riders. The restaurant is responsible for balancing economic concerns with the delivery of orders safely and efficiently [48].

## VII. METHODOLOGY

### A. Research Design

In contrast to centralized systems, which allow a small number of businesses to control the entire industry and manage their delivery staff and dining establishments, this study utilized a decentralized system based on blockchain technology to achieve superior results in delivering food to consumers' homes.

### B. Data Collection

The dataset used for the study [49] is the "Food Delivery Dataset," which has been obtained from Kaggle [50], a popular data science competition platform.

### C. Implementation of the Code

Python programming was used to implement the proposed paradigm. Two separate simulations were conducted to demonstrate the superiority of decentralization, one for the centralized and one for the decentralized systems. [These simulations utilized a variety of constants.](#) It was anticipated that 20 restaurants, each with a staff of five riders, would participate in the scenario. These two constants were added together to ascertain the total number of riders. For simplicity of computation, all riders were assumed to travel at 40 km/h. In addition, an average rider rating (*ARR*) of 2.5 is considered, and each rider was assigned an average rating of 2500, expressed in thousands. The Manhattan distance metric was utilized to calculate the distance between two locations because it is simple and practical [51] [52].

$$manhattan = |x_1 - x_2| + |y_1 - y_2| \quad (1)$$

where  $(x_1, y_1)$  and  $(x_2, y_2)$  being the two points. The rider purchasing procedure of the auction system adheres to predetermined guidelines. To provide a reasonable upper limit, the utmost proposal allowed for a rider is 20% of the total order cost. To protect the interests of the rider, however, a minimum bid requirement of 10% of the total order cost is established. As a result, a rider is guaranteed at least 10% of the total cost of the order as remuneration for their assistance with super delivery.

The total distance traveled by the rider is the aggregate of the distances from the rider's starting point to the restaurant and from the restaurant to the customer's destination. For the sake of simplicity, it is essential to assume that the food will be available for the rider to pick up and deliver to the designated location as soon as the order is placed.

As described in the reference [53] (*Pelane et al., 2016*), the Elo Rating System maintains and updates rider ratings. [A modified form of the Elo rating system has been used](#)

[because the original Elo system is used when two players are involved, but here, only one person is involved, the rider. So, to incorporate it, the two parameters taken are \*averageRiderRating\* or \*ARR\* and the \*olderRating\* or \*OR\*.](#) The following contains formulas for the Elo rating system.

$$E = \frac{1}{1 + 10^{\frac{ARR - OR}{400}}} \quad (2)$$

In the above equation, *ARR* represents *averageRiderRating*, and *OR* represents *olderRating*. By the above equation, *E* is calculated, which will be further used to calculate the change that is observed in the rider rating. A simple algorithm based on the Elo rating system is used to calculate the change. If the customer rating is greater than  $0.8 * average\_customer\_rating$ , one is subtracted from *E*; otherwise, 0 is subtracted from *E*. This value is multiplied by 32 to calculate the change observed in the rider rating. This change is then added to the older rating of the rider, which gives the new rating of the rider. Since the rider's rating has been updated, the average customer and average rider ratings have also been updated so that the database remains up-to-date.

$$ACR = \frac{ACR * ordersDone + customerRating}{ordersDone + 1} \quad (3)$$

$$ARR = \frac{numberBoys * ARR + change}{numberBoys} \quad (4)$$

In the above equations, *ACR* represents *averageCustomerRating*, *ordersDone* represents the number of orders done so far by the rider, *customerRating* indicates the rating that the customer gave to the rider, *change* represents the change in the rider rating, which is calculated above, and *numberBoys* represents the total number of riders present.

[In summary, the rider's rating is calculated using the Elo Rating system. In the Elo system, a particular \*E\* is calculated using the average rider rating and older rating, and then this \*E\* is used to calculate the average customer and average rider ratings.](#)

All the values used in the simulations are normalized values. [The min-max normalization](#) technique is used to normalize the values present in the dataset [54] [55]. The formula for the same is given below.

$$X_r = \frac{X - Min}{Max - Min} \quad (5)$$

Where  $X_r$ ,  $X$ ,  $Min$ , and  $Max$  represent the normalized distance of the rider  $r$ , the actual distance of the rider  $r$ , the distance of the closest rider, and the distance of the farthest rider from the restaurant respectively.

### D. Specifications of Decentralized Simulations

This section describes the method for resolving issues with dynamic pricing [56] and maximizing overall profitability for riders and restaurants. In this arrangement, restaurants provide the least expensive delivery fee first. Riders can then submit

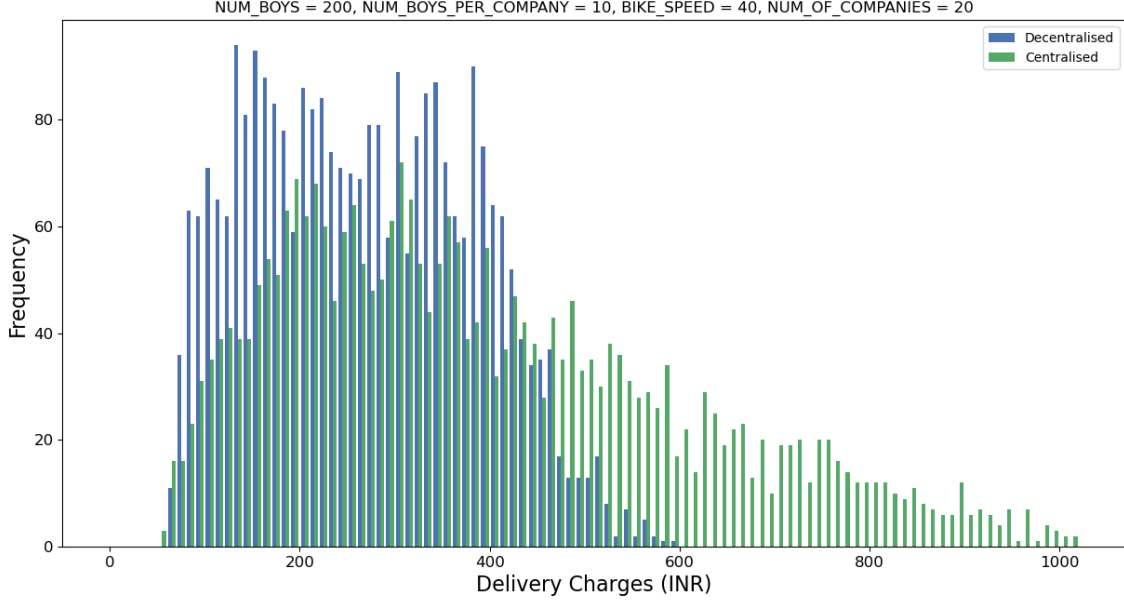


Fig. 3: Histogram of Delivery Charges in Centralized and Decentralized System

their proposals, commonly called pricing, to complete the order. This bidding procedure optimizes delivery costs while encouraging rider competition [57].

By permitting riders to submit bids for delivery, the strategy attempts to discover the most cost-effective option while preserving service quality. By offering competitive pricing, riders are incentivized to compete for the order, which may lead to a more efficient distribution of orders [58]. This dynamic pricing strategy ensures delivery expenses are minimized based on market conditions and the availability of riders.

In addition, this bidding mechanism assists restaurants by allowing them to receive competitive proposals from riders. They may choose an offer reflecting their delivery cost expectations, maximizing their profitability.

This strategy promotes a dynamic, competitive market where consumers and restaurants benefit from the dynamic pricing and purchasing system. It reduces delivery costs, ensures restaurants receive the highest possible payment for each order, and allows riders to compete and win charges based on their aggressive offers [59].

The array of riders is indicated by  $\Gamma_R = (R_j | j \in \mathbb{A}), \mathbb{A} = \{0, 1, 2, \dots, A\}$ . The array of restaurants is indicated by  $\Gamma_C = (C_i | i \in \mathbb{M}), \mathbb{M} = \{0, 1, 2, \dots, M\}$  and the array of orders is indicated by  $\Gamma_O = (O_l | l \in \mathbb{Z}), \mathbb{Z} = \{0, 1, 2, \dots, Z\}$ . Our proposed model considers a variety of variables, including time, bid, order cost, and customer rating. The k-nearest neighbors technique is used to calculate the expected arrival time of the rider at the restaurant. The bid factor includes the three auction phases in which the k-nearest riders willing to partake bid for delivering the food order. Cost is significant because it influences the rider's decision and strongly correlates with client contentment.

These two factors are very significant. If the order cost is high, the restaurant will prioritize selecting a highly-rated rider to deliver the food order quickly and safely. In contrast, if the order cost is modest, the restaurant may offer the order to a rider with a lower rating.

After receiving the minimum and maximum bid values for completing the order, the riders anxious to participate in the auction may begin bidding. The standard rate of commission is 10%. **Our strategy employs three bidding rounds to select the most cost-effective passenger for delivery.** After three rounds of bargaining, the rider with the best combination of a favorable rating based on the order cost and a reduced price is chosen. The selected rider is then tasked with completing the order's delivery in the shortest period possible. Let  $T_{ki}$ ,  $C_{ki}$ ,  $M_{ki}$  and  $V_i$  denote the proposed time, cost of the order, bid, and customer rating, respectively, with the  $i^{th}$  rider for the  $k^{th}$  order where  $i \in \mathbb{M}, \mathbb{M} = \{0, 1, 2, \dots, M\}$ .

We will provide each rider with their machine-predicted bid, determined using a simple formula obtained through experimentation. The formula used is

$$MPB = \lceil \frac{0.1 * order * 0.5 * distance}{40} \rceil + (0.1 * order) \quad (6)$$

A machine-predicted bid ( $MPB$ ) is calculated using the cost of the order ( $order$ ) and the total distance that the rider will travel to complete the order ( $distance$ ).

### VIII. AUCTIONING MODEL

Each rider has a unique score  $v_i$ , which is calculated by their consumer rating and order history for different orders  $\Gamma_O = (O_k | k \in \mathbb{N}), \mathbb{N} = \{0, 1, 2, \dots, N\}$ .

The proposal for the rider must fall between the minimum and maximum bid amounts for the application to be approved.

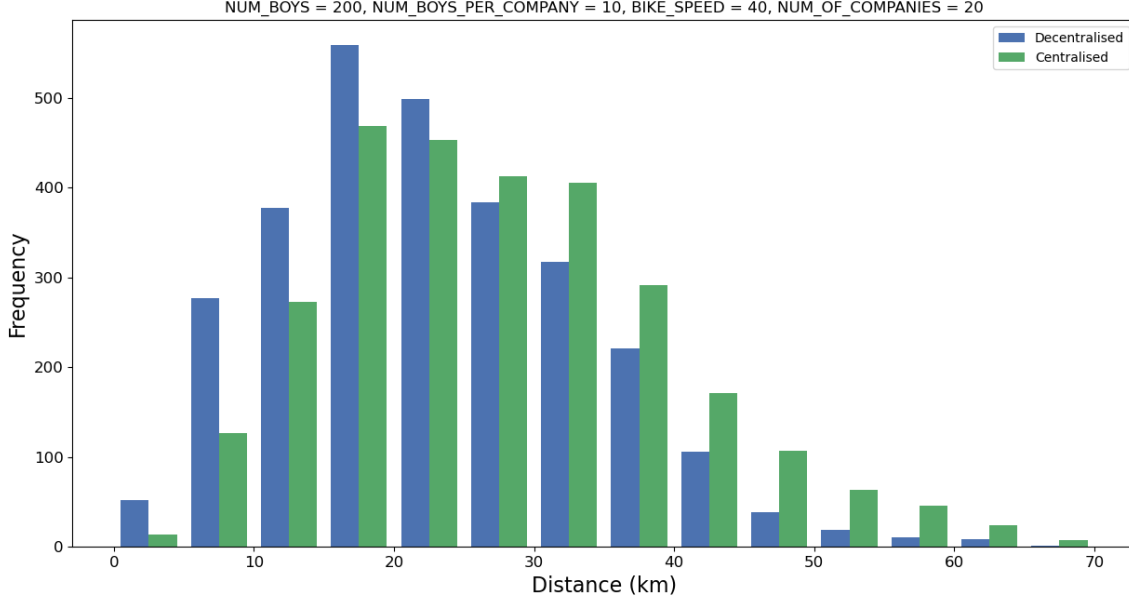


Fig. 4: Histogram of Distance Travelled in Centralized and Decentralized System

This is how our suggested method operates. All participating riders are informed of crucial information, such as the pricing of the order, the consumer's position, and the restaurant's address. This allows any rider not interested in the auction to withdraw if necessary. The number of bidders for the order will be displayed on the application. The identities and bids of riders will remain confidential, and neither the auction winner nor their winning bid will be disclosed on the application. In our use case, we anticipate participation from the  $k$  nearest riders.

Customers can rate the riders on a scale from 1 to 5 following each order. This grading system significantly impacts the precision of machine-predicted bids matching riders' bids. A rider's consumer rating will be amended following each order to reflect the customer's experience and level of satisfaction with the rider's service.

The rating gets updated when the rider's bid  $i$  gets accepted. The restaurant is shown both the riders' proposals and the machine's predicted value. The restaurant chooses the rider based on the total price of the order. The device also recommends the most suitable rider among the participants. Despite their reduced cost, the restaurant knows assigning a high-priced order to a low-rated rider would be imprudent. Therefore, the restaurant provides high-value orders to riders with high ratings and low-value rankings to riders with low ratings.

## IX. NUMERICAL ANALYSIS

Several use cases were executed to evaluate our proposed strategy. In each case, a customer orders food from a restaurant. After receiving the order, the restaurant stores the order details on its server, including the items, client location, and restaurant location. Our system identifies the

closest rider and notifies them of the order details. The riders are also provided a minimum and maximum bid range within which their proposal must lie. In addition, they receive a machine-predicted bid produced by the algorithm above.

After obtaining the machine-predicted bid, the riders participate in bidding. The auction has three stages, allowing riders to decrease or increase their offers gradually. The restaurant evaluates the bids by factoring in the scores derived from rider ratings and past conduct. The establishment ultimately decides which rider will be notified as the auction victor and responsible for delivering the order. The auction process is conducted three times, and the number three was obtained through experimentation. It was observed that running an auction thrice gave the best results, and hence, the model output also increased.

When the rider arrives at the client's location, the client provides them with a One-Time Password (OTP) and inquires about their experience. According to this rating, the server updates the rider's historical activity, calculating the rider's average rating. Riders should carefully consider their bids in future auctions, as increasing their prices could result in a loss. Riders must submit their ultimate proposal price during the auction's conclusion.

## X. RESULTS

We have developed and deployed a blockchain-based application enabling consumers to place real-time restaurant orders. Participating restaurants submit their lowest and most competitive bids for each order in this program. The closest available riders are notified promptly and given a brief window to submit their proposals. These messages to riders contain vital information such as the restaurant's address, the delivery



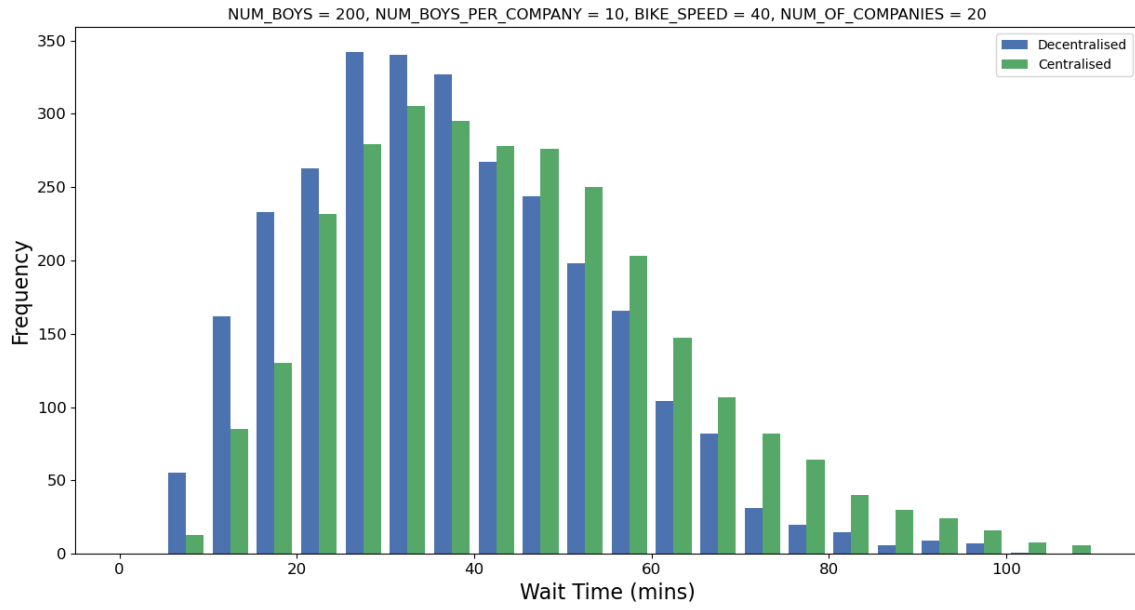


Fig. 5: Histogram of Total Waiting Time in Centralized and Decentralized System



Fig. 6: Comparison of Cumulative Wait time in Centralized and Decentralized System

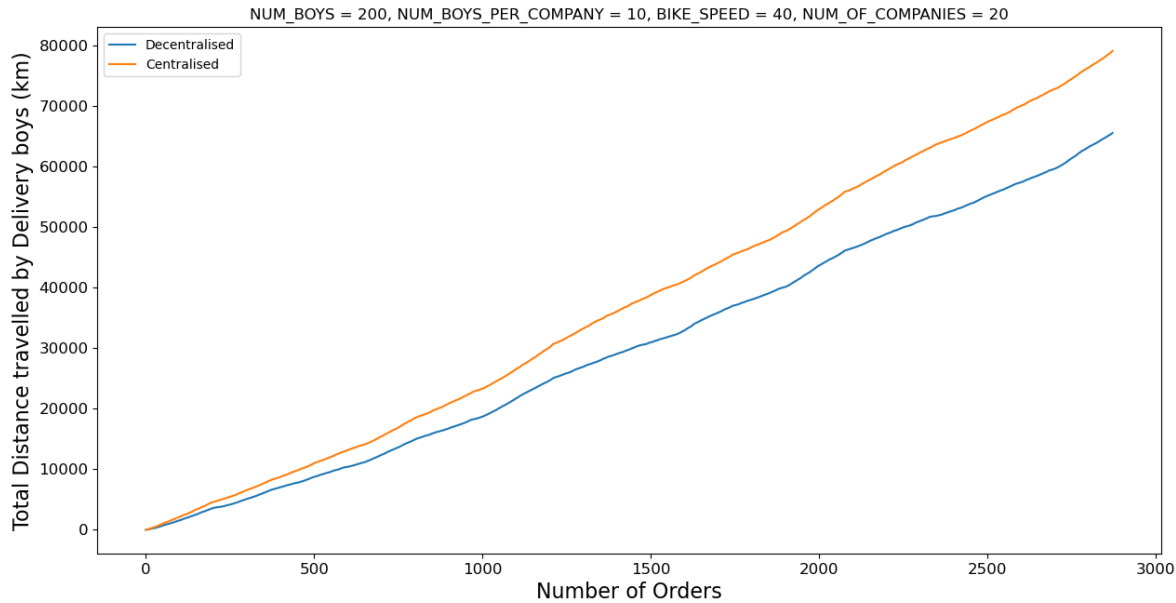


Fig. 7: Comparison of Cumulative Distance Travelled in Centralized and Decentralized System

location, the minimum and maximum bid limits established by the restaurant, and a machine-predicted bid.

After receiving these notices, riders can submit proposals for the orders in question. The restaurant then conducts three distinct auctions in which all riders' bids are aggregated. The restaurant must choose only one rider based on the bids received and the rider's current rating, presented after the auction.

After analyzing the observed patterns in distinct orders, the following figures explain our conclusions.

In the food block application context, Figure 3 compares the delivery costs of a centralized and decentralized system. The investigation revealed a rapid and significant decline in shipping costs. This finding suggests that the new decentralized strategy will likely attract a broader clientele, boosting sales. To provide exact figures, it is intriguing that the decentralized approach results in an average delivery fee of 267.20 units. In comparison, the centralized method produces an average delivery price of 283.25 units.

Figure 4 provides a comprehensive analysis of the total distance riders traveled, compelling the case for a significant decrease. This decrease is attributable to the ability of the decentralized system to effectively distribute services among riders in the restaurant's vicinity, thereby reducing the need to contact a rider from a distance. As a result, this optimization of rider proximity not only saves time for both commuters and consumers but also increases the overall efficiency of the service.

This time savings allows riders to complete more orders, increasing operational effectiveness. Additionally, it concurrently increases customer satisfaction significantly. To place this improvement in context, it is essential to observe that the average distance traveled in the centralized system is

26.62 kilometers. In contrast, in the decentralized system, it is only 22.23 kilometers. This drastically reduces the distance traveled by 16.49 percent.

Figure 5 compares total wait times and demonstrates that wait times have decreased substantially in the decentralized system. This modification will significantly enhance the user experience as a whole.

To be precise, the centralized system's average rider delay time is 43.86 minutes. In contrast, the decentralized system has drastically decreased this average wait time to 37.08 minutes, a remarkable 15.47% decrease. This significant reduction in wait times highlights the efficiency gains enabled by the decentralized strategy.

In Figures 6 and 7, a cumulative examination of delay times and travel distance for both the centralized and decentralized systems reveals significant decreases in both parameters. These results demonstrate the enormous gains in efficiency made possible by the decentralized system.

The delay time has decreased by an astounding 15.47%, while the total distance traveled has reduced by an astounding 16.49%. These reductions demonstrate the efficacy and benefits of the decentralized system, which ultimately enhances operational performance and customer satisfaction.

Figure 8 depicts a comprehensive analysis of customer-submitted evaluations of riders. Notably, there is a discernible and commendable upward trend in riders' ratings within the decentralized system. This pattern has numerous beneficial effects. First, the riders' earning potential is enhanced, as a higher rating enables them to charge more for their services. Second, it promotes competition among riders for orders and rating advancements. This competitive dynamic facilitates the continuous development and longevity of the online supper delivery system.

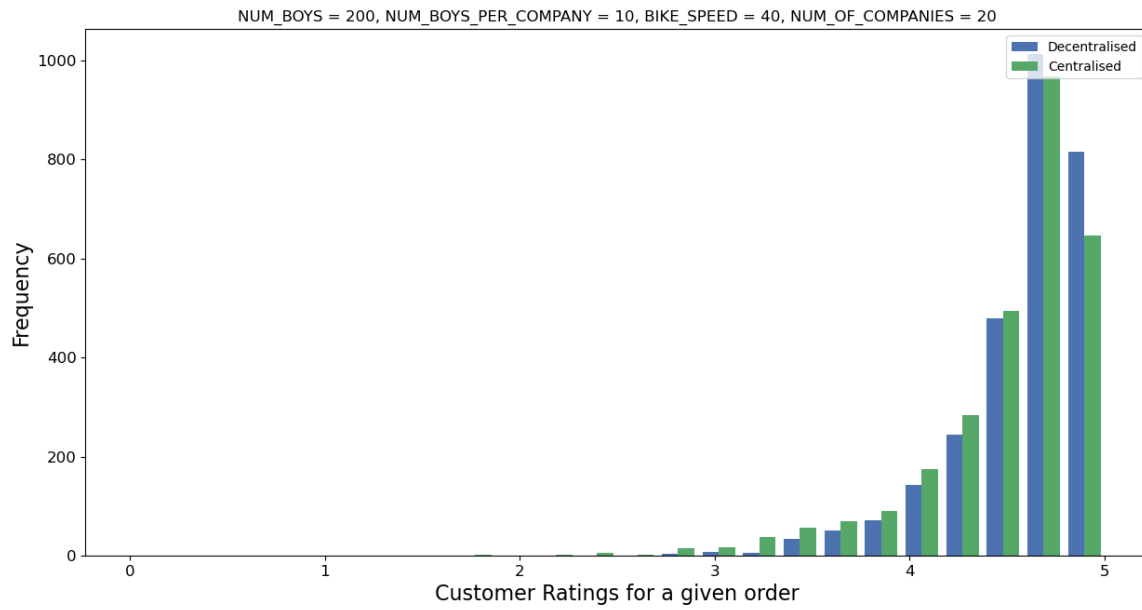


Fig. 8: Histogram of Rating given by the Customer in Centralized and Decentralized System

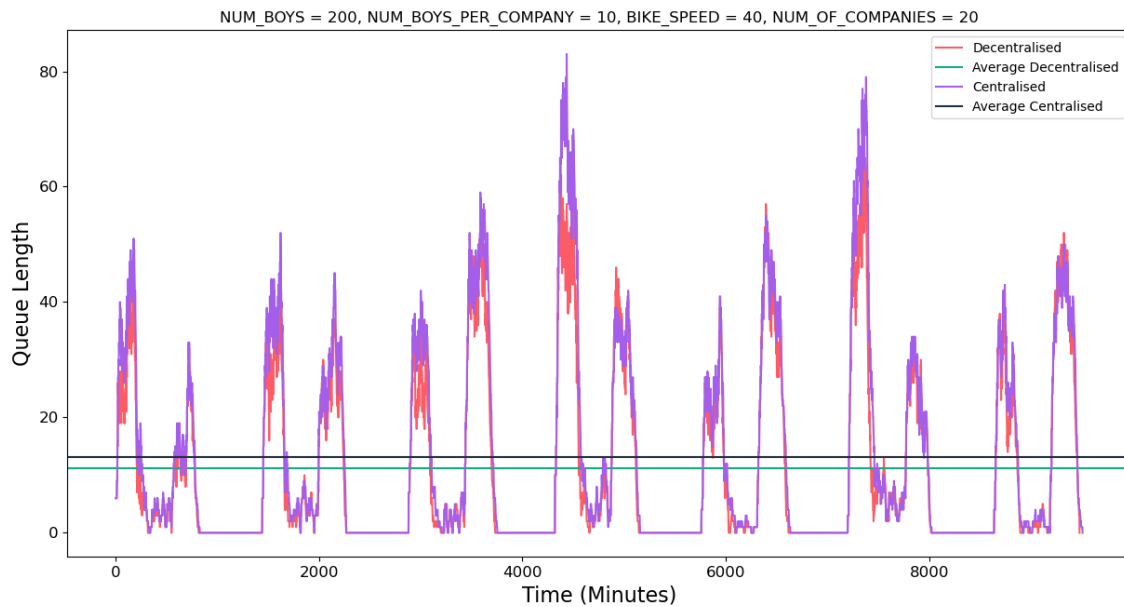


Fig. 9: Analysis of Queue Length in Centralized and Decentralized System

Comparing numerical data reveals that the average customer rating given to riders in the centralized system is 4.39, which is quantifiable. Under the decentralized system, motorcyclists receive an average rating of 4.57, representing a remarkable increase of 4.04% in rider ratings. This improvement highlights the positive effects of the decentralized system on rider efficiency and the system's overall efficacy.

Examining the wait times in both systems, as depicted in Figure 9, reveals a significant reduction in wait times within the decentralized system. This reduction in delay times has a significant impact on the operational efficiency.

The average delay time for the centralized system is estimated to be 142.66 minutes. The decentralized system reduces the average wait time to 124.25 minutes, 87.09% of the queue duration under the conventional, centralized approach.

Moreover, restaurants have made substantial savings after a protracted simulation period. The astounding total savings amount to 79,608.4169 rupees. The reduced travel distance of the decentralized system is responsible for a significant portion of these savings, which total 33,509.4169 Rs. These price reductions illustrate the financial gains and operational efficiencies made possible by the decentralized system.

In the following results, it has been established that the decentralized delivery system is better than the centralized system of delivering goods. One of the primary reasons for this is that in a decentralized system, no companies are dominating the market; instead, everyone works independently through a decentralized system. Hence, there is no competition among the companies and no scope for exploitation, which the companies can do to the riders. Secondly, the auction system decides the riders' rates, which allows them to choose at what price they want to go, in contrast to the previous system, where the company dominated the riders' prices. They had to abide by the rules and regulations of the company forcibly.

## XI. FUTURE RESEARCH DIRECTIONS

The following study presents the idea of shifting from a centralized system to a decentralized food delivery system. This research paper assumes that the rider carries only one order at a time to implement the idea. However, in theory, it has been hinted that it can also be expanded to carrying multiple orders simultaneously. So, one of the significant future works could be to expand the implementation of this research work from one order delivery to a multiple order delivery problem, where a rider can collect multiple orders at a time and then deliver them to the respective customers.

## XII. CONCLUSION

Within our proposed framework, dining establishments establish the boundaries for order pricing by setting minimum and maximum thresholds. At the same time, delivery personnel submit bids that form the foundation of an auction-oriented system. A rating system calculates the most suitable rider for each order. Furthermore, integrating blockchain technology in the online meal ordering business is expected to enhance restaurant profitability and augment rider earnings compared

to the current system. Additionally, our technological solution effectively tackles the issue of delayed food delivery and mitigates fraudulent activities via a one-time password (OTP) method, guaranteeing secure transactions. As stated in the results section, the proposed model demonstrates superior performance compared to the traditional system.

## REFERENCES

- [1] M. Nofer, P. Gomber, O. Hinz, and D. Schiereck, "Blockchain," *Business & Information Systems Engineering*, vol. 59, pp. 183–187, 2017.
- [2] V. Krishna, *Auction theory*. Academic press, 2009.
- [3] C. Li, M. Miroso, and P. Bremer, "Review of online food delivery platforms and their impacts on sustainability," *Sustainability*, vol. 12, no. 14, p. 5528, 2020.
- [4] R. Zhang, R. Xue, and L. Liu, "Security and privacy on blockchain," *ACM Computing Surveys (CSUR)*, vol. 52, no. 3, pp. 1–34, 2019.
- [5] V. Hassija, G. Bansal, V. Chamola, V. Saxena, and B. Sikdar, "Blockcom: A blockchain based commerce model for smart communities using auction mechanism," in *2019 IEEE International Conference on Communications Workshops (ICC Workshops)*, 2019.
- [6] A. Muthoo, *Bargaining theory with applications*. Cambridge University Press, 1999.
- [7] D. Scifo, M. Scifo, D. Scifo, L. Valente, and R. Valente, "Online food ordering system and method," Aug. 13 2009, uS Patent App. 12/069,895.
- [8] Z. He, G. Han, T. Cheng, B. Fan, and J. Dong, "Evolutionary food quality and location strategies for restaurants in competitive online-to-offline food ordering and delivery markets: An agent-based approach," *International Journal of Production Economics*, vol. 215, pp. 61–72, 2019.
- [9] J. J. Cotter, "Food order/delivery system," Jan. 10 1989, uS Patent 4,797,818.
- [10] S. E. Kimes, "The current state of online food ordering in the us restaurant industry," *Cornell Hospitality Report*, vol. 11, no. 17, pp. 6–18, 2011.
- [11] K. Christidis and M. Devetsikiotis, "Blockchains and smart contracts for the internet of things," *Ieee Access*, vol. 4, pp. 2292–2303, 2016.
- [12] V. Hassija, M. Zaid, G. Singh, A. Srivastava, and V. Saxena, "Cryptober: A blockchain-based secure and cost-optimal car rental platform," in *2019 Twelfth International Conference on Contemporary Computing (IC3)*. IEEE, 2019, pp. 1–6.
- [13] V. Hassija, V. Chamola, and S. Zeadally, "Bitfund: A blockchain-based crowd funding platform for future smart and connected nation," *Sustainable Cities and Society*, p. 102145, 2020.
- [14] N. Jain, T. Agrawal, P. Goyal, and V. Hassija, "A blockchain-based distributed network for secure credit scoring," in *2019 5th International Conference on Signal Processing, Computing and Control (ISPCC)*. IEEE, 2019, pp. 306–312.
- [15] V. Hassija, G. Bansal, V. Chamola, N. Kumar, and M. Guizani, "Secure lending: Blockchain and prospect theory-based decentralized credit scoring model," *IEEE Transactions on Network Science and Engineering*, 2020.
- [16] K. Huang, X. Zhang, Y. Mu, X. Wang, G. Yang, X. Du, F. Rezaeiabagha, Q. Xia, and M. Guizani, "Building redactable consortium blockchain for industrial internet-of-things," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 6, pp. 3670–3679, 2019.
- [17] —, "Building redactable consortium blockchain for industrial internet-of-things," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 6, pp. 3670–3679, 2019.
- [18] —, "Building redactable consortium blockchain for industrial internet-of-things," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 6, pp. 3670–3679, 2019.
- [19] —, "Building redactable consortium blockchain for industrial internet-of-things," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 6, pp. 3670–3679, 2019.
- [20] F. Tian, "An agri-food supply chain traceability system for china based on rfid & blockchain technology," in *2016 13th international conference on service systems and service management (ICSSSM)*. IEEE, 2016, pp. 1–6.
- [21] M. Mettler, "Blockchain technology in healthcare: The revolution starts here," in *2016 IEEE 18th International Conference on e-Health Networking, Applications and Services (Healthcom)*. IEEE, 2016, pp. 1–3.

- [22] K. Biswas and V. Muthukumarasamy, "Securing smart cities using blockchain technology," in *2016 IEEE 18th international conference on high performance computing and communications; IEEE 14th international conference on smart city; IEEE 2nd international conference on data science and systems (HPCC/SmartCity/DSS)*. IEEE, 2016, pp. 1392–1393.
- [23] V. Hassija, V. Chamola, V. Saxena, D. Jain, P. Goyal, and B. Sikdar, "A survey on iot security: Application areas, security threats, and solution architectures," *IEEE Access*, 2019.
- [24] S. Huh, S. Cho, and S. Kim, "Managing iot devices using blockchain platform," in *2017 19th international conference on advanced communication technology (ICACT)*. IEEE, 2017, pp. 464–467.
- [25] A. Yadav, D. Yadav, S. Gupta, D. Kumar, and P. Kumar, "Online food court payment system using blockchain technology," in *2018 5th IEEE Uttar Pradesh Section International Conference on Electrical, Electronics and Computer Engineering (UPCON)*. IEEE, 2018, pp. 1–7.
- [26] V. Hassija, G. Bansal, V. Chamola, V. Saxena, and B. Sikdar, "Blockcom: A blockchain based commerce model for smart communities using auction mechanism," in *2019 IEEE International Conference on Communications Workshops (ICC Workshops)*. IEEE, 2019, pp. 1–6.
- [27] V. Hassija, V. Chamola, S. Garg, N. G. K. Dara, G. Kaddoum, and D. N. K. Jayakody, "A blockchain-based framework for lightweight data sharing and energy trading in v2g network," *IEEE Transactions on Vehicular Technology*, 2020.
- [28] B. Segendorf, "What is bitcoin," *SverigesRiksbankEconomicReview*, vol. 2014, pp. 2–71, 2014.
- [29] G. Bansal, V. Hassija, V. Chamola, N. Kumar, and M. Guizani, "Smart stock exchange market: A secure predictive decentralized model," in *2019 IEEE Global Communications Conference (GLOBECOM)*. IEEE, 2019, pp. 1–6.
- [30] L. T. Chai and D. N. C. Yat, "Online food delivery services: Making food delivery the new normal," *Journal of Marketing advances and Practices*, vol. 1, no. 1, pp. 62–77, 2019.
- [31] N. Chandrasekhar, S. Gupta, and N. Nanda, "Food delivery services and customer preference: a comparative analysis," *Journal of Foodservice Business Research*, vol. 22, no. 4, pp. 375–386, 2019.
- [32] N. Haller, C. Metz, P. Nesser, and M. Straw, "A one-time password system," Tech. Rep., 1998.
- [33] Y. S. Lee, H. Lim, and H. Lee, "A study on efficient otp generation using stream cipher with random digit," in *2010 the 12th international conference on advanced communication technology (ICACT)*, vol. 2. IEEE, 2010, pp. 1670–1675.
- [34] L. E. Peterson, "K-nearest neighbor," *Scholarpedia*, vol. 4, no. 2, p. 1883, 2009.
- [35] R. B. Myerson, "Optimal auction design," *Mathematics of operations research*, vol. 6, no. 1, pp. 58–73, 1981.
- [36] W. Huang, X. Wang, J. Zhang, J. Xia, and X. Zhang, "Improvement of blueberry freshness prediction based on machine learning and multi-source sensing in the cold chain logistics," *Food Control*, vol. 145, p. 109496, 2023.
- [37] J. Xu, Z. Yang, Z. Wang, J. Li, and X. Zhang, "Flexible sensing enabled packaging performance optimization system (fs-ppos) for lamb loss reduction control in e-commerce supply chain," *Food Control*, vol. 145, p. 109394, 2023.
- [38] O. Kramer and O. Kramer, "K-nearest neighbors," *Dimensionality reduction with unsupervised nearest neighbors*, pp. 13–23, 2013.
- [39] M. Gates, "Bidding strategies and probabilities," *Journal of the Construction Division*, vol. 93, no. 1, pp. 75–110, 1967.
- [40] Q.-K. Li, H. Lin, X. Tan, and S. Du, "H consensus for multiagent-based supply chain systems under switching topology and uncertain demands," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 50, no. 12, pp. 4905–4918, 2018.
- [41] S. Zhang, X. Li, M. Zong, X. Zhu, and D. Cheng, "Learning k for knn classification," *ACM Trans. Intell. Syst. Technol.*, vol. 8, no. 3, jan 2017. [Online]. Available: <https://doi.org/10.1145/2990508>
- [42] L. Friedman, "A competitive-bidding strategy," *Operations research*, vol. 4, no. 1, pp. 104–112, 1956.
- [43] H. Zhang, X. Zuo, B. Sun, B. Wei, J. Fu, and X. Xiao, "Fuzzy-pid-based atmosphere packaging gas distribution system for fresh food," *Applied Sciences*, vol. 13, no. 4, p. 2674, 2023.
- [44] M. Curran, "A scientific approach to bidding: Range estimating," *Constructor*, vol. 62, no. 1, 1975.
- [45] J. Yang, K. Yang, Z. Xiao, H. Jiang, S. Xu, and S. Dustdar, "Improving commute experience for private car users via blockchain-enabled multitask learning," *IEEE Internet of Things Journal*, 2023.
- [46] X. Xu, C. Wang, and P. Zhou, "Gvrp considered oil-gas recovery in refined oil distribution: From an environmental perspective," *International Journal of Production Economics*, vol. 235, p. 108078, 2021.
- [47] E. Triantaphyllou and E. Triantaphyllou, *Multi-criteria decision making methods*. Springer, 2000.
- [48] R. Seamans and F. Zhu, "Repositioning and cost-cutting: The impact of competition on platform strategies," *Strategy Science*, vol. 2, no. 2, pp. 83–99, 2017.
- [49] "Food delivery dataset | kaggle," <https://www.kaggle.com/datasets/gauravmalik26/food-delivery-dataset?select=train.csv>, (Accessed on 08/30/2023).
- [50] "Kaggle: Your machine learning and data science community," <https://www.kaggle.com/>, (Accessed on 08/30/2023).
- [51] M. Malkauthekar, "Analysis of euclidean distance and manhattan distance measure in face recognition," in *Third International Conference on Computational Intelligence and Information Technology (CIIT 2013)*. IET, 2013, pp. 503–507.
- [52] S. Sai, U. Mittal, V. Chamola, K. Huang, I. Spinelli, S. Scardapane, Z. Tan, and A. Hussain, "Machine un-learning: An overview of techniques, applications, and future directions," *Cognitive Computation*, pp. 1–25, 2023.
- [53] R. Pelánek, "Applications of the elo rating system in adaptive educational systems," *Computers & Education*, vol. 98, pp. 169–179, 2016.
- [54] L. Al Shalabi, Z. Shaaban, and B. Kasasbeh, "Data mining: A preprocessing engine," *Journal of Computer Science*, vol. 2, no. 9, pp. 735–739, 2006.
- [55] V. Chamola, V. Hassija, A. Singh, U. Mittal, R. Pareek, P. Mangal, D. Devika, M. Mahmud, and D. Brown, "Metaverse for education: Developments, challenges and future direction," 2023.
- [56] A. V. Den Boer, "Dynamic pricing and learning: historical origins, current research, and new directions," *Surveys in operations research and management science*, vol. 20, no. 1, pp. 1–18, 2015.
- [57] X. Xu, Z. Lin, X. Li, C. Shang, and Q. Shen, "Multi-objective robust optimisation model for mdvrpls in refined oil distribution," *International Journal of Production Research*, vol. 60, no. 22, pp. 6772–6792, 2022.
- [58] D. E. Griffith and R. T. Rust, "The price of competitiveness in competitive pricing," *Journal of the Academy of Marketing Science*, vol. 25, pp. 109–116, 1997.
- [59] N. Caldwell, H. Walker, C. Harland, L. Knight, J. Zheng, and T. Wakeley, "Promoting competitive markets: The role of public procurement," *Journal of Purchasing and supply management*, vol. 11, no. 5-6, pp. 242–251, 2005.